

Linear transformations

For linear transformation matrix.

$$[T] = [T(e_1) \quad T(e_2) \quad \dots \quad T(e_n)].$$

Each column represent the transformation of that specific base e_i so a complete matrix covers all space n in $\mathbb{R}^n$.

A fn $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if & only if there exists a matrix $[T] \in M_{m,n}$ such that

$$T(v) = [T]v \quad \text{for all } v \in \mathbb{R}^n$$

A fn of LT can be written as a matrix.

eg

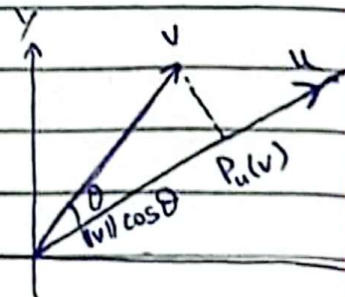
$$T(v_1, v_2, v_3) = (3v_1 - v_2 + v_3, 2v_1 + 4v_2 - 2v_3)$$

$$[T] = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 4 & -2 \end{bmatrix}$$

Projection

u is a unit vector.

$[P_u] = uu^T$ is a projection matrix



So when we need to find projection on vector u

Projected vector = $[P_u]v \rightarrow$ vector to be projected

reflection through
line $y=mx$

$$F_L = \frac{1}{m^2+1} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$$

reflection through
given inclination θ , $y=x$

$$F_\theta = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$

Reflection

Standard matrix of reflection $[F_u] = 2uu^T - I$

It is basically reflection about vector u .

Rotation

The standard matrix of rotation in 2D plane is

$$R^\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Rotation in higher dimension 3D has matrices.

$$[R_{yz}^\theta] = [R_{yz}^\theta(e_1) \ R_{yz}^\theta(e_2) \ R_{yz}^\theta(e_3)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

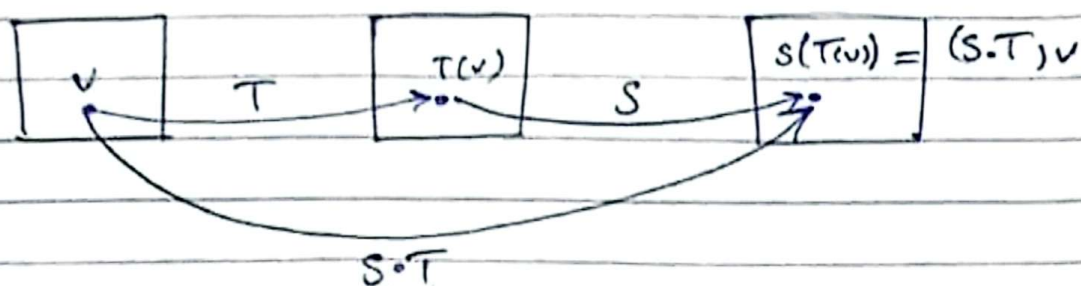
rotation along x-axis.

$$[R_{zx}^\theta] = \begin{bmatrix} \cos(\theta) & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}, [R_{xy}^\theta] = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Around a particular line (3D) [around a unit vector]

$$R_u^\theta = (\cos \theta)I + (1 - \cos \theta)uu^T + \sin \theta \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}$$

Composition of Linear transformation



Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ & $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations.
Then composition is fn. $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$

$$(S \circ T)(v) = S(T(v)) \quad \forall v \in \mathbb{R}^n$$

T is a standard matrix $[T] \in M_{m,n}$ & $[S] \in M_{p,m}$
then $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ is a L.T & its standard matrix is

$$[S \circ T] = [S][T]$$

The order of multiplication is that last transformation multiplies first & then it moves backwards.